

VENKATESHWARLU SARANGI, Ph.D

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PROFESSIONAL SUMMARY

AI/ML Research Specialist with Ph.D. in Physics from City University of Hong Kong and HKUST, backed by 10+ years in research and applied data science. Expert in machine learning and deep learning foundations, including mainstream algorithms like CNN, RNN, Transformer, and GAN, with hands-on proficiency in computer vision (CV), natural language processing (NLP), large models, and reinforcement learning (RL). Skilled in TensorFlow, PyTorch, Python, and C++ for distributed training, model compression, and edge inference; proven in integrating AI with vertical industries like healthcare and AI for Science. Contributed to open-source via GitHub, focusing on scalable MLOps for production-grade large models (e.g., LLMs, multimodal models) and interdisciplinary collaborations.

TECHNICAL SKILLS

Machine Learning, Deep Learning, CNN, RNN, Transformer, GAN, Computer Vision (CV), Natural Language Processing (NLP), Large Models, Reinforcement Learning (RL), TensorFlow, PyTorch, Python, C++, Distributed Training, Model Compression, Edge Inference, MLOps, LLM Fine-Tuning, RAG Systems, Vector Databases (FAISS, Chroma), Prompt Engineering, Knowledge Distillation, AI Integration (Healthcare, AI for Science), Open-Source Contributions.

DOMAIN SPECIFIC SMALL LANGUAGE MODEL OPTIMIZATION & DEPLOYMENT

Capstone Project-1 (March, 2026)

- Production-Grade Knowledge Distillation (KD) Pipeline for Small Language Models (SLMs).
- Transferred knowledge from large teacher (Qwen2.5-7B-Instruct) to compact student (Qwen2.5-1.5B-Instruct) model, enhancing reasoning and planning via SFT, reward modeling, and RL methods like PPO and DPO
- Achieved comparable performance with 80% reduced inference compute (latency/memory), while synthesizing large-scale, high-quality data through rewriting, augmentation, and generation for pretraining, SFT, and RL stages.
- Production deployment: vLLM serving (OpenAI-compatible API), Docker containerization, Streamlit; implemented robust evaluation methodologies to assess model performance.
- Key metrics: Perplexity/BLEU convergence, TTFT (Time-To-First-Token), throughput vs. baseline teacher; integrated tool usage and agent capabilities for complex task solving.

Capstone Project-2 (April, 2026)

- Production-Grade Vision Language Model (VLM) Training & Serving Platform.
- Built multimodal SLMs for Computer Vision + NLP Applications, incorporating RL for foundation model enhancement and system 2 thinking with advanced decoding like MCTS and A.
- Delivered full-stack VLM platform for training/serving small-scale vision-language models, teaching models to use tools, interact with APIs, and build multi-agent systems for complex tasks.
- Production infrastructure: vLLM multi-modal serving (Qwen-VL/LLaVA), tensor parallelism, FP8 inference on H100; Docker + K8s orchestration.
- Key capabilities: Image-text alignment, zero-shot classification, cross-modal retrieval with CLIP-style embeddings; applied data synthesis and evaluation for pretraining, SFT, and RL.

WORK EXPERIENCE

DATA SCIENTIST & ML ENGINEER

Freelancer at XRadio Ltd, HK
(Aug, 2025 – March, 2026)

Deep Learning and Multimodal Systems for ECG Signal Intelligence

- Built an explainable deep learning model for 12-lead ECG **time-series data classification**, addressing challenges from non-stationary signals and inter-patient variability.
- Evaluated multiple deep learning and vision architectures including ResNet34, ResNet50, DenseNet, VGG16, Vision Optimized Transformers (ViT) and Vision-Language Models (VLMs), multimodal AI systems for healthcare prediction tasks.
- Achieved strong model performance with ResNet34, reaching **AUROC 0.98 and F1-score 0.826** across nine arrhythmia categories.

DATA SCIENTIST & RESEARCH ENGINEER

Gense Technologies Ltd, HK
(Oct, 2022 – Aug, 2025)

Projects & Technical Experience

- Designed and deployed end-to-end supervised learning pipelines with ML models for clinical prediction tasks, achieving 87% sensitivity and >99% specificity on imbalanced biomedical datasets.
- Engineered feature extraction and signal processing workflows using Python (NumPy, Pandas, Scikit-learn) to improve SNR and diagnostic quality in time-series bio-conductivity data.

- Achieved a 100% improvement in skin electrode contact impedance without conductive gels, improving signal sensitivity.
- Developed and validated signal enhancement and impedance analysis models for ECG and EIT datasets, improving signal-to-noise ratio and diagnostic reliability.
- Built automated MLOps workflows with Docker, CI/CD, and MLflow for experiment tracking, model versioning, and reproducible deployments across cloud environments. Applied statistical validation frameworks (k-fold cross-validation, ROC-AUC optimization, hypothesis testing) to ensure diagnostic-grade model reliability and regulatory compliance.
- Developed synthetic data augmentation strategies (SMOTE, bootstrapping) to address limited clinical datasets, improving model generalization by 20%.
- Proficient in Python, SQL, PyTorch, Scikit-learn, and cloud ML platforms (GCP, Firebase, AWS SageMaker) for scalable production systems.

DATA SCIENTIST AND COMPUTATIONAL ANALYST

Anvipro IT Solutions, India

(December, 2021 – June, 2022)

Materials Science & Computational Modeling

- Applied supervised learning and statistical modeling techniques to analyze material properties, identify trends, and support predictive insights.
- Conducted physics-based simulation and data analysis using COMSOL Multiphysics to study material flexibility (>90%), stress distribution, hardness, and structural behavior.
- Performed materials data preprocessing, cleaning, and refinement using MATLAB and Python basic libraries to improve analysis quality and model readiness.
- Improved simulation and analytical workflows through iterative model refinement, results evaluation, and reliability analysis for engineering and materials-focused projects.
- **Technologies:** Python, Data processing, Feature Engineering, EDA, NumPy, Pandas, Scikit-learn, OriginLab, SQL, Power BI, & Statistical Modeling.

PH.D RESEARCH SCHOLAR

CityU HK

(August, 2015 – October 2021)

Materials Engineering / Computational Modeling / Data Analysis

- Architected piezoelectric electrodes and integrated AI-driven computational modeling for material structure analysis using X-ray and neutron diffraction datasets, aligning with predictive AI solutions for operational efficiency.
- Delivered Rietveld refinement and data processing, feature engineering, and regression-based parameter optimization, fine-tuning methods, hyperparameter tuning to extract structural insights from complex datasets, incorporating Machine Learning frameworks like Scikit-learn for enhanced prediction tasks.
- Published research in Nature Communications, Communications Physics, Physical Review B, Scientific Reports, and other high-impact journals, focusing on innovative AI applications in materials science.

JOURNAL PUBLICATIONS

- S. Venkateshwarlu et al., **Nature Communications Physics**, (2020).
- S. Venkateshwarlu et al., **Journal of Materials Research**, Cambridge University Press JMR-0830 (2019).
- S. Nayak, S. Venkateshwarlu, et al., **Journal of the American Ceramic Society**, (2019).
- Frederick, Sarangi Venkateshwarlu, et al., **Chemistry of Materials** (2021).
- A. Pramanick, S. Venkateshwarlu, et al., **Physical Review B**, (2021).
- Liang, Zhuoxin Liu, S. Venkateshwarlu, et al., **Nature, Light: Science & Applications**, (2021).
- G. Srinivas, S. Venkateshwarlu, et al., **Applied Physics Letters**, (2015).
- S. Venkateshwarlu, et al., **Journal of Modern Materials**, (2016).
- Pramanick, Venkateshwarlu, **Journal of the European Ceramic Society**, (2023).
- Nirmal and Sarangi Venkateshwarlu et al., **arXiv:2508.10940**, (2025).
- Kwok, W.C., Sarangi Venkateshwarlu et al., **Nature Scientific Reports** (2025).

EDUCATION

IIT Bombay India and CityU HK,

- **Ph.D.** Materials Science and Eng (Physics), City University of Hong Kong, (Oct, 2021)
- **M.Tech.** Materials Science and Eng (Physics), IIT Bombay, India, (June, 2013)

TRAINING | CERTIFICATIONS

IIT Bombay India and CityU HK,

- **Generative AI with Large Language Models (LLMs)**
- DeepLearning.AI & Amazon Web Services | Coursera | February 2026

- Skills: Transformers Architecture (FLAN-T5, BERT, GPT, DeepSeek, Qwen, Phi-2, Gemma) · RL models · PyTorch · PEFT · Fine-tuning · LoRA · QLoRa · Knowledge Distillation(KD) · RAG · LangChain · LangGraph · Deployment · vLLM · OpenWebUI · FastAPI · Streamlit · RunPOD · LightningAI · HuggingFace, · Optuna Optimization
- **Machine Learning Specialization** | August 2025
- **Deep Learning and NLP Specialization** | January 2026
- Stanford University & DeepLearning.AI | Coursera
- **End-to-End MLOPS Bootcamp** | Udemy | April 2026-(On-going)
- **AWS Certified Cloud Practitioner** | Udemy | April 2026-(On-going)